Nicholas (Nick) Wagner

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Education

Ph.D. 2024 **Baylor University**
Dissertation: Deformation of the Martian Lithosphere Through Time Advisor: Dr. Peter James

B.Sc. 2019 **Colorado School of Mines**
Major: Geophysics
Minor: Computer Science Specializing in Data Science

Work Experience

2024-Present **Brown University**
Postdoctoral Research Associate

2019-2024 **Baylor University**
Research & Teaching Assistant

2018-2019 **EDCON-PRJ**
Data Processing Technician

Research Experience

2023 **NASA/JPL Planetary Science Summer School**
Telecomms Chair, Gravity Science Lead *New Frontiers Class Titan Orbiter Concept*

2022 **InSight**
Student Observer of STM25 *InSightSeer Program*

2021 **Keck Institute for Space Studies Workshop**
Study Participant *Next-Generation Planetary Geodesy*

2021 **NASA/JPL**
Research Intern

2019 **United States Geological Survey**
Student Contractor

2018 **Lunar and Planetary Institute**
Research Intern

2018 - 2019 **Colorado School of Mines**
Undergraduate Research Fellow

Publications

In Prep

- **Wagner, N.**, Lau, H., Berne, A., Frankenberg, N., Matsuyama, I. Geodetic Evidence for a Compositionally Distinct Lunar Nearside Mantle
- Manuscript is currently in review with coauthors. Estimated time of submission: within 1 month.

- **Wagner, N.**, James, P., McGovern, P. The Argument for a Geodynamically Controlled Emplacement of Pavonis Fossae

- Manuscript is written, but some analysis will need to be redone. Estimated time of submission: within 6 months.

- Jindal, A., **Wagner, N.**, Trussell, A., O'Rourke, J., Head, J., Kerr, M., Hayes, A., Birch, S. Investigating Global Patterns of Tectonic Deformation in Venusian Canali
- Project led by Abhinav Jindal. I am contributing expertise on lithospheric flexure and stress modeling and contributing to the discussion. Analysis is finished, and the manuscript is currently being written. Estimated time of submission: within 2 months.

- North, A., James, P., **Wagner, N.**, Moriarty, D. Geophysical Constraints on Bulk Density at the Gruithuisen Domes
- Project led by Allie North. I am contributing expertise on gravity modeling and Lunar volcanism. Manuscript is currently in revisions amongst coauthors. Estimated time of submission: within 3 months.

In Review

- Berne, A., **Wagner, N.**, Matsuyama, I., Goossens, S., Cheng, K., Genova, A., Roviera-Navarro M., Bagheri, A., Qin, C., Marusiak, A., Hemingway, D., Lau, H., Zhong, S., Nimmo, F. Molten Root for Mars's Crustal Dichotomy Inferred from Seasonal Tidal Tomography *Nature*
- Project led by Alex Berne. I am contributing expertise on polar cap and atmospheric effects to the time-variable gravity signal of Mars. I am also contributing to the interpretation of our results in terms of the geodynamic state of Mars. Manuscript is currently in review at *Nature*.

2026

- Broad, K., Sadler, B., James, P., Hoover, S., **Wagner, N.**, Hood, D. A geophysical survey of the Kentland Crater formation *Geosciences*
- Project led by Katie Broad. I helped provide field work for collecting magnetotelluric data. *MDPI Geosciences* `https://doi.org/10.3390/geosciences16040155`

2025

- **Wagner, N.** & James, P. New Geophysical Constraints for Intrusive Magmatism at Large Martian Volcanoes: Implications for Crustal Thickness and Volatile Outgassing *Journal of Geophysical Research: Planets* `https://doi.org/10.1029/2025JE008959`

2024

- Seltzer, C., Bott, K., Brouwer, G., Burtt, D., Gentgen, C., Lien, R., Abbate, J., Head, T., Hill, J., Gandarillas, V., Green, A., Larson, J., Montiel, N., Moreno, R., Mullikin, E., Radzom, B., **Wagner, N.**, Wijesekara, P., Nash, A., Keane, J. THUNDER: A Titan Orbiter New Frontiers Mission Concept *Planetary Science Journal*. `https://doi.org/10.3847/PSJ/ad973e` - I contributed expertise related to measuring the gravity field of Titan, and it would tell us about the state of formation of the moon.

- **Wagner, N.**, James, P., Ermakov, A., Sori, M. Evaluating the Use of Seasonal Surface Displacements and Time-Variable Gravity to Constrain the Interior of Mars *Journal of Geophysical Research: Planets*. `https://doi.org/10.1029/2023JE008053`

2022

- Xiao, H., Stark, A., Schmidt, F., Hao, J., Steinbrügge, G., **Wagner, N.**, Su, S., Cheng, Y., Oberst, J. Spatio-temporal level variations of the Martian Seasonal North Polar Cap from co-registration of MOLA profiles *Journal of Geophysical Research: Planets*. `https://doi.org/10.1029/2021JE007158`
- I contributed expertise in regards to how measured variations in the polar caps of Mars can induce periodic deformations of the surface.

Awards and Fellowships

Graduate School Travel Award **Baylor Graduate School**
2020-2024

Graduate School Fellow **Baylor Graduate School Fellowship**
2019-2024

Outstanding Teaching Assistant **Baylor Geosciences Department**
2022

Texas Space Grant Fellow **Texas Space Grant Consortium**
2020

Teaching Experience

2019 - 2024 **Baylor University**
Teaching Assistant

GEO1401: Earthquakes and Natural Hazards
GEO3319: Intro to Geophysics
GEO5328: Geodynamics

2019 **Colorado School of Mines**
Teaching Assistant

GPGN268: Geophysical Data Analysis

Conference Abstracts (†indicates oral presentation)

2025

- †**Wagner**, N., Lau, H., Berne, A., Frankenberg, N. *The Lopsided Moon: Tidal Signals of a Heterogeneous Interior* (AGU)

- Frankenberg, N., **Wagner**, N., Lau, H. *The geochemical context of geophysically derived nearside-farside Lunar mantle variations* (AGU)

- Wieczorek, M., Alvarez-Candal, M., Angrisani, M., Attree, N., Bhattacharya, A., Blewett, D., Broquet, A., Burkhard, L., Cascioli, G., Collins, G., Cone, K., Crameri, F., Denton, A., Elkins-Tanton, L., Fa, W., Filiberto, J., Flahaut, J., Frigeri, A., Guimond, C., Guray Hatipo, Y., Jacobson, S., Kim, D., Lucas, A., Maia, J., Mandon, L., Martinot, M., Miljkovic, K., Montesi, L., Nawal, A., Oran, R., Pan, L., Park, R., Putzig, N., Rodriguez, S., Roos-Serote, M., Root, B., Ruedas, T., Schmidt, F., Snape, J., Tai Udovicic, C., Tosi, N., van der Wal, W., **Wagner, N.** *Planetary Research - A Diamond Open Access Journal For Planetary Science* (LPSC)

- **Wagner, N.**, James, P. *Geophysical Evidence for High I/E Ratios at Large Martian Volcanoes* (LPSC)

2024

- **Wagner, N.**, James, P., Ermakov, A., Sori, M. *Using Seasonal Surface Displacements and Time-Variable Gravity to Constrain the Interior of Mars* (Mars Geophysics After InSight)

- North, A., James, P., **Wagner, N.**, Moriarty, D., Kiefer, W., Siegler, M. *Crustal Context of the Moon's Gruithuisen Domes* (LPSC)

- †**Wagner, N.**, James, P. *Higher Intrusive Magma Volumes Found at Olympus Mons from Gravity and Topography Data* (LPSC)

2023

- **Wagner, N.**, James, P., Ermakov, A., Sori, M. *New Estimates of Seasonal Surface Displacements and Time Variable Gravity on Mars* (LPSC)

- North, A., James, P., **Wagner, N.**, Moriarty, D. *Gravity Constraints on Bulk Density and Igneous Intrusion at Gruithuisen Domes* (LPSC)

- Moriarty, D., James, P., North, A., **Wagner, N.** *Evidence for Widespread Silicic Lithologies Across the Gruithuisen-Mairan Region* (LPSC)

- Morris, J., Schurmeier, L., **Wagner, N.**, Baker, S., Bill, C., Mohanna, S., Roth, N., Sanderson, H., Sridhar, S., Woodley, S., Daubar, I., Fernando, B., Newman, C., Panning, M. *InSightSeers: Peering into Invited Student Participation of a Mission Science Team Meeting* (LPSC)

2022

- **Wagner, N.**, James, P., Ermakov, A., Sori, M. *Quantifying Lithospheric Deflection Caused by Seasonal Mass Transport from the Polar Layered Deposits on Mars* (LPSC)

- **Wagner, N.**, Park, R., James, P. *A Geophysical Investigation of the Compensation State of Hellas Planitia* (LPSC)

- Sori, M., Bramson, A., Byrne, S., James, P., Ojha, L., **Wagner, N.** *Gravity Science Constrains the Presence and Volume of Mid-Latitude Ice Sheets on Mars* (LPSC)

- Keane, J., Sori, M., Ermakov, A., Austin, A., Bapst, J., Berne, A., Bierson, C., Bills, B., Boening, C., Bramson, A., D'Amico, S., Denton, A., Evans, A., Hemingway, D., Hernandez, S., Hogstrom, K., Izquierdo, K., James, P., Johnson, B., Kahre, M., Lau, H., Navarro, T., Neveu, M., Nimmo, F., O'Rourke, J., Ojha, L., Paik, HJ., Park, R., Rosen, P., Simons, M., Smith, D., Smrekar, S., Soderlund, K., Steinbrgge, G., Tikoo, S., Vance, S., **Wagner, N.**, Weber, R., Zebker, H., Zuber, M. *Next-Generation Planetary Geodesy: Results from the 2021 Keck Institute for Space Studies Workshops* (LPSC)

- Sori, M. M., Ermakov, A. I., Keane, J. T., Bierson, C. J., Bills, B. G., Bramson, A. M., D'Amico, S., Evans, A. J., Hemingway, D. J., Izquierdo, K., James, P. B., Johnson, B. C., Kahre, M. A., Navarro, T., O'Rourke, J. G., Ojha, L., Paik, H. J., Park, R. S., Simons, M., Smith, D. E., Smrekar, S. E., Soderlund, K. M., Steinbrgge, G., Tikoo, S. M., Vance, S. D., **Wagner, N. L.**, Weber, R. C., Zebker, H. A. *Next-Generation Planetary Geodesy: Results from the 2021 Keck Institute for Space Studies Workshops* (Low-Cost Science Mission Concepts for Mars Exploration)

2021

- **Wagner, N.**, James, P. *Evaluating Magma Ascent at Pavonis Mons, Mars Using Stress from Flexure* (LPSC)

2019

- **Wagner, N.**, Kay, J., Schenk, P. The Orientation of the Bladed Terrain Feature in Tarturus Dorsa, Pluto and Possible Reorientation of Pluto (LPSC)

2018

- **Wagner, N.**, Kay, J., Schenk, P. Study of the Orientation of the Bladed Terrain Feature in Tartarus Dorsa, Pluto (AGU)

Mentorship

Neal Frankenberg **Brown Undergraduate**
Thermodynamic and geochemical interior modeling of the lunar interior: a geophysical perspective *2025-2026*

Allie North — **Baylor Undergraduate**
Gravity survey of silicic volcanism on the Moon — *2022-2024*

Skylar Hoover — **Baylor Undergraduate**
Magnetotelluric survey of Kentland Crater, Indiana — *2022-2023*

Outreach & Seminars

AstroAssembly — **October 2025**
Is the Moon Made of Parmesan or Brie?
How Tides Tell Us What's Inside the Moon — *Seagrave Observatory*

Astronomy on Tap — **June 2025**
Is the Moon Made of Parmesan or Brie?
How Tides Tell Us What's Inside the Moon — *Moniker Brewing*

Lunch Bunch — **April 2025**
Tidal Tomographic Insights into Lunar Interior Structure:
Is the Moon Made of Parmesan or Brie? the Moon — *Brown University*

Solid-Earth Seminar — **October 2024**
Martian Geophysics and Geodesy:
Insights into Volcanism and Planetary Structure — *Georgia Tech*

Special Geodesy Seminar — **July 2024**
Next Generation Martian Geodesy: What Could The
Seasonal Polar Caps Tell Us About The Martian Interior? — *NASA Goddard*

Total Solar Eclipse — **April 2024**
Mars: Dead or alive? — *Baylor University*

Organizations and Competitions

2025-2026 — **Planetary Lunch Bunch Organizer**
Brown University

2020 - 2023 — **Graduate Student Association (GSA)**
Geoscience Representative (Baylor University)

2017 - 2019 — **Undergraduate Student Government (USG)**
Senate Representative (Colorado School of Mines)

2017 - 2018 — **2018 NASA Rasc-Al Competition**
Deputy Project Lead and Quality Control Lead

Presented at AIAA 2018: Paper number AIAA-2018-5108. `https://doi.org/10.2514/6.2018-5108`

2018 — **Colorado School of Mines Geophysics Summer Field Camp**
Final Report Lead

Led the completion of a 200-page report detailing the findings of the field camps surveys in Pagosa Springs, CO.

2018 — **Student Society of Geophysicists (SSG)**
Treasurer

Scientific Service

2025 — **NASA Artemis IV LSSW**
Participant

2021-2025 — **NASA Grant Review Panels**
Executive Secretary and Panelist — *ROSES*

2025 — **Science: Advances**
Journal Reviewer

2025 — **AGU: Geophysical Research Letters**
Journal Reviewer

2022-2025 — **AGU: JGR-Planets**
Journal Reviewer

2023-2025 — **Elsevier: Icarus**
Journal Reviewer